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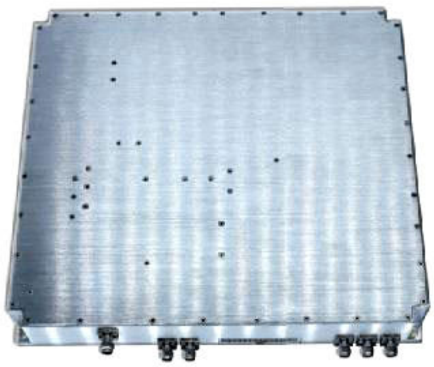
9kHz to 20GHz Signal Source Module

Ultra Wideband Signal Source Module



Description:

Corech Microwave's Ultra-Wideband Signal Source Module operates from 9kHz to 20GHz/ 40GHz, featured with ultra wideband, CW/Pulse mode, ultra low noise of -105dBc/Hz@10kHz , low spurious and low consumption. They are very rugged and stable to be integrated in various systems.

PN:

PN	Description	Option
ST-9K20G		

Key Specifications:

- Frequency Range: 9kHz to 20GHz
- Frequency Steps: 0.001Hz
- Output Power: 0dBm ~ +15dBm
- Setting Time: 300us
- Spurious: 60dBc
- Phase Noise: -105dBc/Hz@10kHz at 20GHz
- External Reference: 10MHz
- Internal Reference: 100MHz
- RF Ouput Waveform: Sine wave / pulse modulation
- External Pulse Input: LVTTL

Plot no. 845, DSIIDC Industrial Area KHASRA NO. 33/23 (0-7) & 37/4 (0-10)Gali no. 6, near Harnandi dharam kanta Mundka Industrial Area, New Delhi, Delhi, 110041

• Electrical Specifications

External Reference (50Ω)

Parameter	Min.	Typ.	Max.	Note
Input Frequency		10MHz		
Input Power	0dBm		+5dBm	
Phase Noise			-150dBc/Hz@1kHz	

Internal Reference (50Ω)

Parameter	Min.	Typ.	Max.	Note
Frequency		100MHz		
Frequency stability		0.1ppm		
Temperature stability		0.1ppm		
Phase Noise			-155dBc/Hz@1kHz	

Reference Output

Parameter	Min.	Typ.	Max.	Note
Output Frequency	10MHz, 100MHz			
Output Power	0dBm			10MHz
	0dBm			100MHz
Phase Noise			-140dBc/Hz@1kHz	10MHz
			-145dBc/Hz@1kHz	100MHz

9kHz to 20GHz Signal Source Module

Dimensions:

Table 1: J30J-15ZKP Interface Definition

J30J Pin	Definition	J30J Pin	Definition
1	+24V	9	T+
2	+24V	10	T-
3	+24V	11	R+
4	+24V	12	R-
5	GND	13	NC
6	GND	14	NC
7	GND	15	NC
8	GND		

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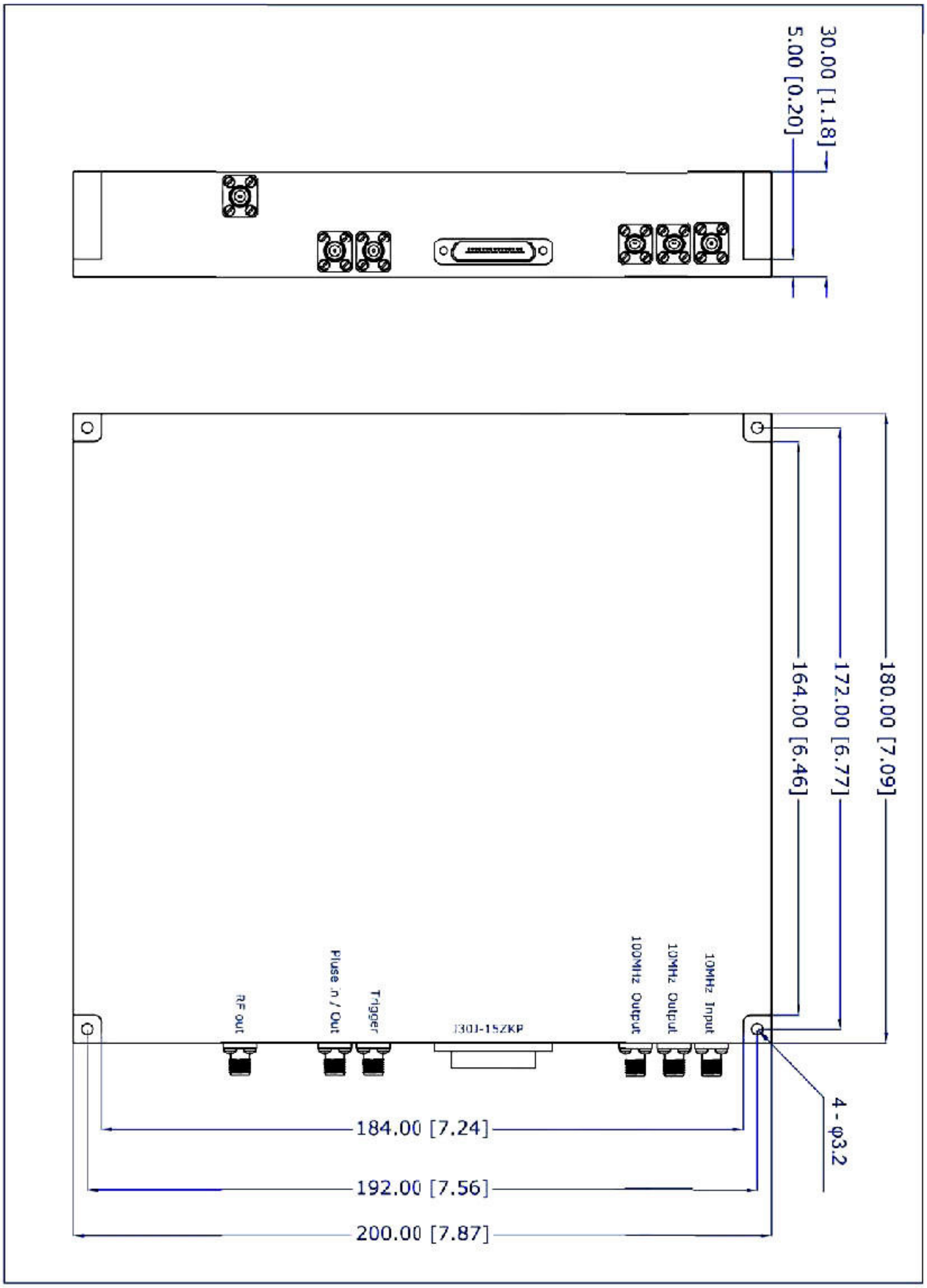
9kHz to 20GHz Signal Source Module

- **Dimensions:**



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- **Option 2 （Sweep）**

Sweep

Parameters	Min.	Typ.	Max.
Operating Mode		Step Sweep	
Sweep Range	9kHz		20GHz
Dwell Tine	10ms		10s
Time Resolution		1us	
Switching Speed		300us typ. @ Internal Source	

- **Generel Specifications**

Parameters	Min.	Typ.	Max.
Supply Current (Vcc=+24V)			2A
Communication		RS422, Baud Rate: 1Mbps *	
RF Out Connector		2.92mm-Female	
Reference Input/Output		SMA-Female	
Synchronization Signal		SMA-Female	
Power and Control Interface		J30J-15ZP	
Operating Temperature		-20 ~ +40 ℃	
Storage Temperature		-40 ~ +70 ℃	

* Communication protocol will be discussed and agreed upon subsequently

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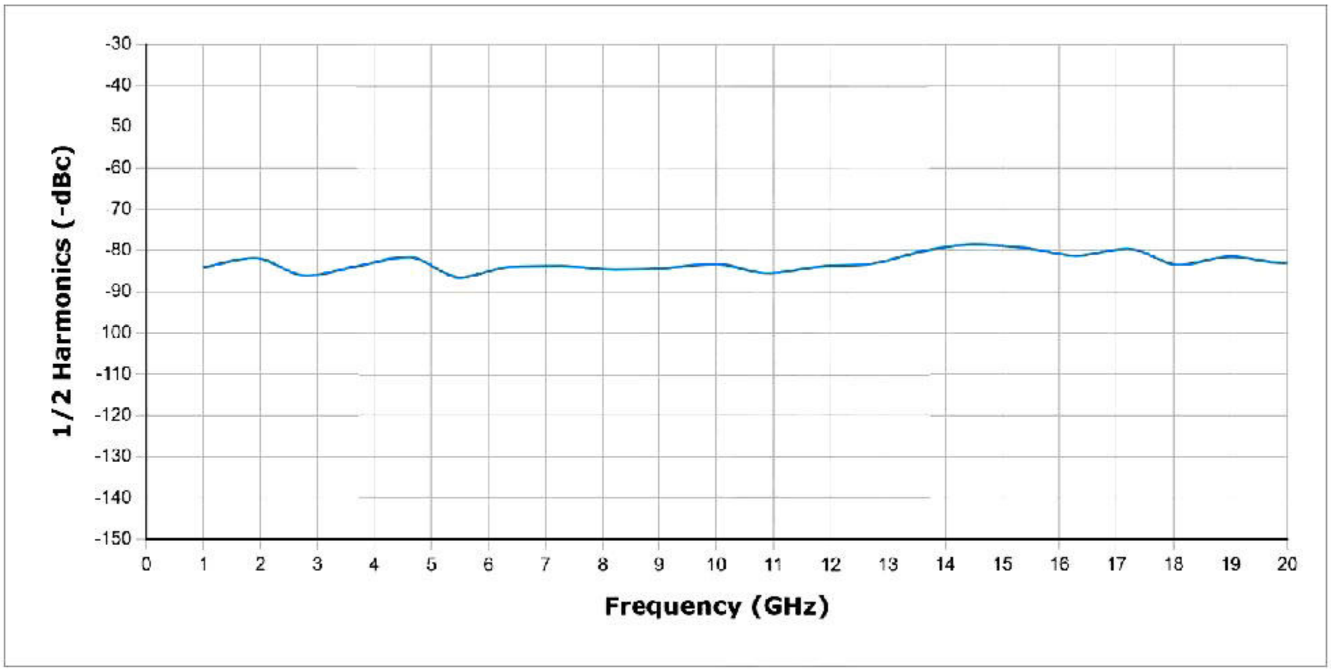
1/2 Harmonics (Typ.)



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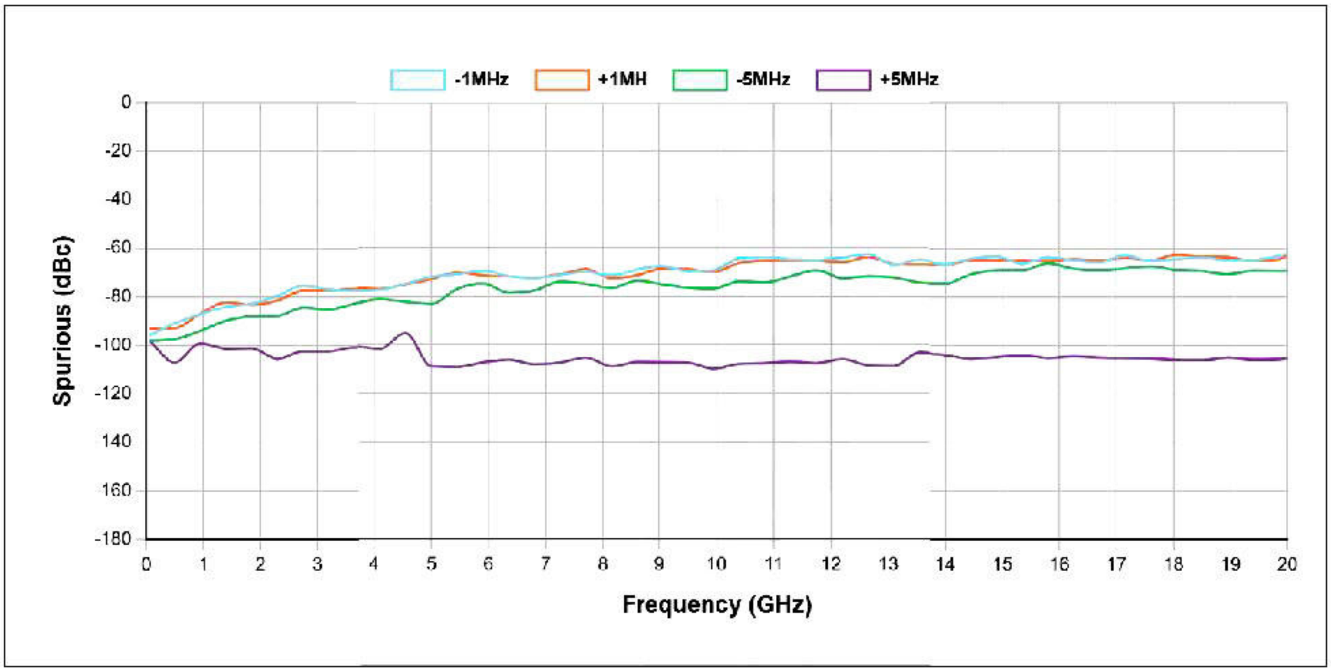
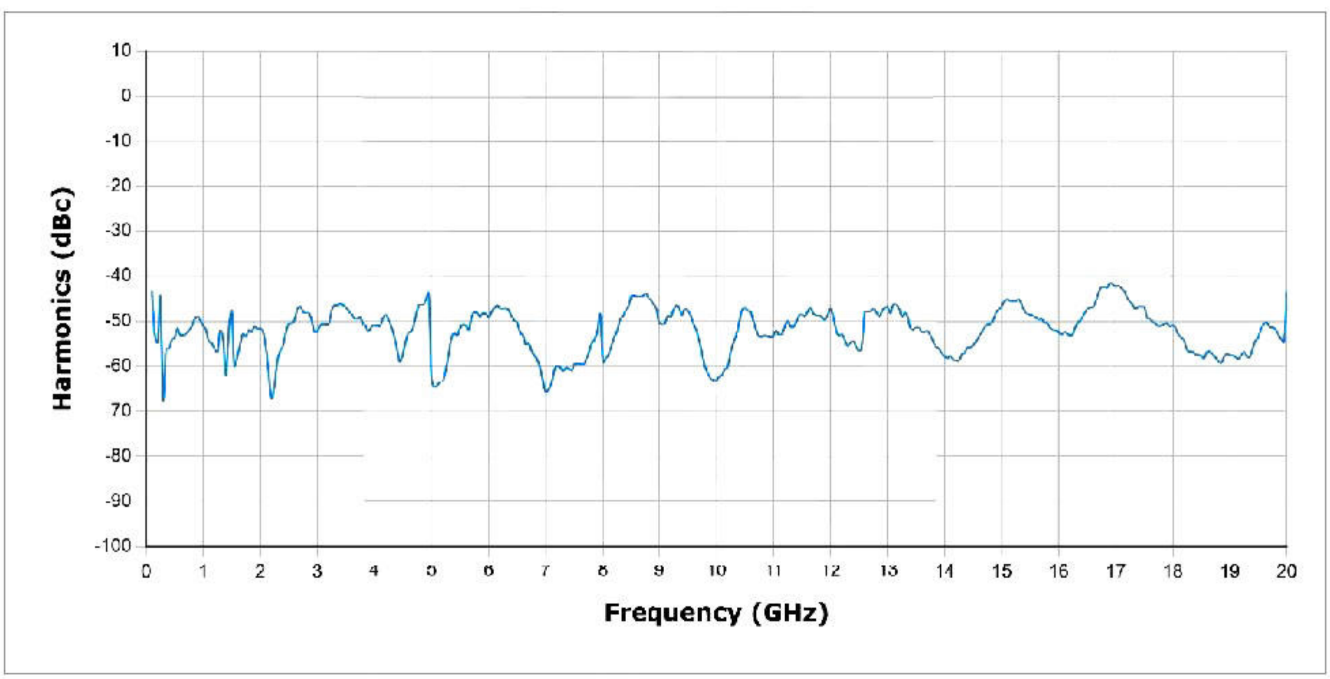
Spurious (Typ.)



Harmonics (Typ.)



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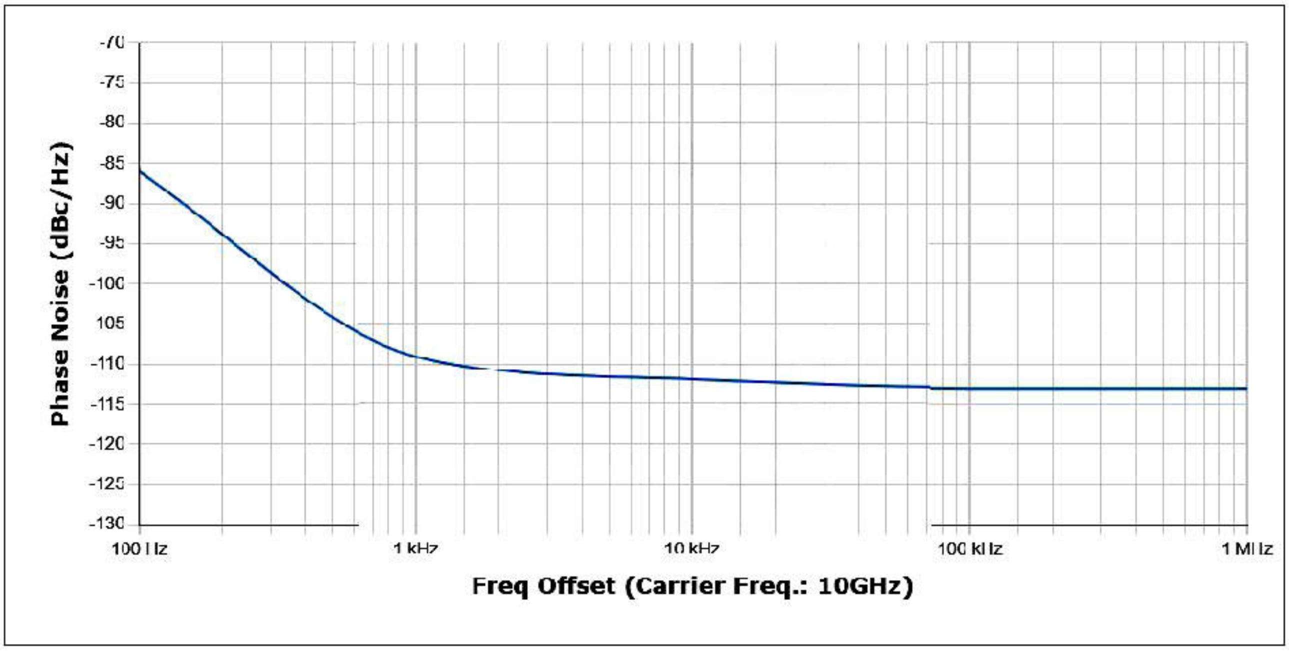
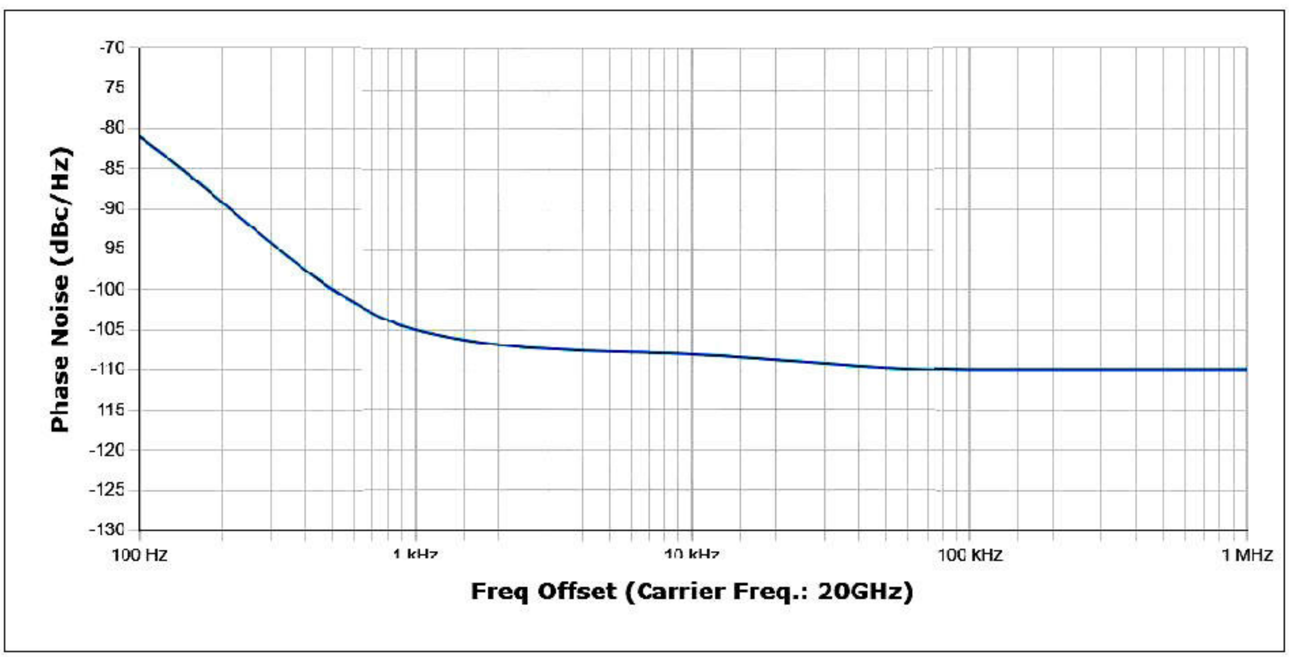
SSB Phase Noise (Typ.)





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• Electrical Specifications

SSB Phase Noise (Type)

Parameters	100Hz	1kHz	10kHz	100kHz	1MHz
dBc/Hz@1GHz	-104	-126	-130	-130	-131
dBc/Hz@10GHz	-85	-106	-110	-110	-111
dBc/Hz@20GHz	-79	-100	-104	-104	-105

• Option 1 （Pulse Generator and Modulation)

Pulse Modulation

Parameters	Min.	Typ.	Max.
On/ Off Ratio		-65dBc	
Min. Pulse Width		100ns	
Min. Period		110ns	
External Pulse Input			
LVTTL		+3.3V	
Internal Pulse Generator			
Square Wave Speed Rate	0.1Hz		10MHz
Period	110ns		100s
Pulse Width	100ns		99.999999s
Rise Time / Fall Time		25ns / 25ns	>100MHz
		150ns / 150ns	≤100MHz
Resolution		100ns	
Delay	20ns		100s

- **Electrical Specifications**

Parameter	Min.	Typ.	Max.	Note
Frequency Range	9kHz		20GHz	
Frequency Steps		0.001Hz		
Output Power	0dBm		+15dBm	25℃
Power Steps			0.5dB	
Output Power Resolution			±2dB	
Setting Time		300us	500us	#1
Harmonics				
1GHz		-47dBc	-40dBc	
10GHz		-53dBc	-40dBc	
20GHz		-50dBc	-40dBc	
1/2 Harmonics				
1GHz		-85dBc	-80dBc	
10GHz		-88dBc	-80dBc	
20GHz		-82dBc	-75dBc	
Spurious (Frequency Offser ≥5kHz)				
1GHz	-85dBc	-80dBc	-60dBc	
10GHz	-80dBc	-75dBc	-60dBc	
20GHz	-74dBc	-70dBc	-60dBc	

#1, Setting Time: 500ns (Excluding communication time)